ES-FA / BTP Local Run Manual

Commands used for the current repo layout

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1 What This Manual Covers

This is the short operating manual for the renamed project layout. The working folders are `p1_training/`, `p2_hardware_model/`, `p3/`, and `output/`. The latest successful smoke run completed the software, analysis, evidence, and output-generation stages. Hardware/Vivado execution was intentionally skipped.

2 Environment Check

```
.\.venv312\Scripts\python.exe -c "import torch, p1_training.training_core, analysis.
    compare; print('ok')"
```

3 Start the Whole Local Smoke Pipeline

```
.\scripts\run_esfa_pipeline.ps1 -PythonExe .\.venv312\Scripts\python.exe -Mode smoke -
    SkipVivado -SkipHardware
```

Successful finish message:

```
ES-FA pipeline complete.
```

4 What the Latest Smoke Run Produced

- Best run: `exp1_hardware_aware_loss`.
- Accuracy: 95.70%.
- Spike sparsity: 56.48%.
- Energy proxy improvement vs baseline: -79.68%.
- Analysis plots: `results/plots/`.
- Evidence tables: `results/analysis/evidence/`.
- Paper sections were regenerated in `output/`: method, experiment, tables, key findings, and the draft paper markdown.

5 Manual Software Commands

```
python p1_training/train_baseline.py
python experiments/run_first2.py
python experiments/run_remaining.py
python iterations/run_all.py
python analysis/compare.py
python analysis/evidence_report.py
python output/generate_paper_output.py
```

6 Hardware Validation Command

Use this when hardware validation is required:

```
python hardware_validation/kv260/scripts/run_hw_validation.py --model-id baseline_paper1
    --scheduler-mode both --clock-mhz 100
```

For simulator-only validation:

```
python hardware_validation/kv260/scripts/run_hw_validation.py --model-id baseline_paper1
    --scheduler-mode both --clock-mhz 100 --skip-vivado
```

7 CLI Demo

```
python deployment/cli_interface/main.py --query "hi"
python deployment/cli_interface/main.py --query "strike␣rate␣167.55␣balls␣39"
python deployment/cli_interface/main.py --query "classify␣results/runtime/sample_signal.
    bin"
```

8 Presentation Reminder

For the current run, present software and analysis results as fresh. Present hardware/Vivado numbers only as existing local validation evidence, not as a fresh board run. The KV260 physical measurement stage is still pending.